

Feixiang Ge

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Focus: LLM Systems | Heterogeneous AI Hardware Inference

Education

University of Chinese Academy of Sciences | M.S. Candidate in Artificial Intelligence Sep. 2025–Present

- Freshman Scholarship, Hangzhou Institute for Advanced Study, UCAS.

Shanghai University of International Business and Economics Sep. 2021–Jun. 2025

B.Eng. in Data Science and Big Data Technology

- Shanghai Municipal and Aijian Special Foundation Scholarships; Outstanding Graduate and Youth League Member.

Industry Experience

Baidu | Model Adaptation and Optimization | vLLM, SGLang, Kunlunxin Feb. 2026–Aug. 2026

- LLM inference optimization:** Profiled and optimized NSA Indexer, MTP/speculative decoding, and context parallelism for DeepSeek-V3.2 on Kunlunxin P800 using SGLang. Improved single-node performance by 20% and PD-disaggregated inference by 50%; identified and resolved speculative-decoding corruption.
- Memory and operator optimization:** Reworked vLLM-Kunlun Workspace Manager memory allocation, doubling the maximum single-node context length for Qwen3-97B (PR 283). Used compute-intensity analysis to replace speculative-attention FP16 kernels with a hardware-suited implementation, $1.8\times$ faster than BF16.
- Framework migration and open-source engineering:** Migrated vLLM-Omni to Kunlunxin from scratch and led multimodal model enablement; major models reached 80% of H20 performance. Ranked 6th of 39 contributors to vLLM-Kunlun with 9 merged PRs spanning refactors, operators, and memory management. Iterated an LLM deployment SOP skill and release agent for automated deployment, performance/accuracy checks, and structured reports.
- Model delivery and distributed inference:** Delivered Qwen3/Qwen3.5, Wan, Qwen-Image, GLM-Image, and Qwen-Image-Edit; enabled smaller-model workloads including Grounding SAM, OpenFold/SAM3, and BEVFormer. Brought up SGLang + HiCache + Dynamo, fixed KV-aware DP-rank and prefix-reuse failures, and improved TPOT by 10%+ while doubling prefix-hit rate on a target dataset.

eBay | AI Platform Engineer | Ray, Kubernetes, Cluster Scheduling Oct. 2024–Aug. 2025

- Resolved multi-resource label scheduling and PyArrow memory-leak issues in Ray Data/Serve; fixes were merged into eBay's internal Ray branch (PR 49124, PR 49756). Built observability and core capabilities for an internal AI-workflow platform.
- Profiled production GPU-cluster utilization and optimized scheduling, memory management, and distributed strategies; fixed a KubeRay scheduling defect that doubled GPU utilization.

Selected Projects

IEEE AICAS 2026 | Efficient VLM Inference and Optimization for AI Chips Rank 6 / 514

- End-to-end VLM acceleration:** Optimized Qwen3-VL-2B-Instruct on PPU-ZW810E for single-batch VQA (1K input / 128 output tokens), reducing TTFT from about 77 ms in vLLM to about 17 ms and increasing throughput from about 220 to 340 tok/s.
- Full-stack optimization:** Designed multi-level image/text KV caching and applied `torch.compile` with multi-strategy Pre-fill/Decode CUDA Graph capture. Implemented fused kernels for QK RMSNorm-RoPE, FlashAttention/FlashDecode, FFN, and KV writes using Triton, CUDA, and TileLang; trace-driven optimization reduced D2H synchronization and small-kernel launches.

Inference Optimization on a RISC-V AI CPU

- **Hardware-aware GEMM optimization:** Used Roofline analysis and PMU profiling to identify VPU-issue and non-matrix-instruction bottlenecks in Qwen3.5-2B. Reordered loops to N-major, switched Q4_1 to Q4_0, expanded quantization scale groups from 32 to 64, and simplified assembly; at 128 tokens, raised Prefill by 289% ($3.89\times$) and Decode by 57% ($1.57\times$) over baseline.
- **Model compression and data-layout optimization:** Reduced the Vision Encoder from 24 to 7 ViT blocks with controlled accuracy loss, doubling vision-prefill throughput. Designed a 64×64 tiled transpose with L1-cache reuse for the SSM convolution-state update, achieving 7–10 $\times$ microbenchmark speedup and roughly 10% end-to-end acceleration.

Research and Publications

- **Dynamical Adapter Fusion: Constructing a Global Adapter for Pre-Trained Model-based Class-Incremental Learning.** RuiQi Liu, Boyu Diao, Zijia An, Runjie Shao, Feixiang Ge, et al. NeurIPS 2026, CCF-A.
- **Using Domain Adapter and Prompt Learner for Few-shot Teaching Action Recognition in Videos.** Feixiang Ge, Xinlei Li, Yan Rong, et al. IJCNN 2025, first author, CCF-C.
- **A Hybrid CNN-Transformer Network for Fine-Grained Tongue Multi-Attribute Classification with Tongue Group Attribute Mask Training in Traditional Chinese Medicine Diagnosis.** Liu J., Ge F., Li F., et al. ICIC 2025, co-first author, CCF-C.
- **General Deep Learning Model Development Assistant.** Software copyright No. 2026SR0303020; first holder.

Technical Expertise

- **LLM serving and distributed inference:** vLLM, SGLang, PD disaggregation, context parallelism, speculative decoding, Dynamo, and HiCache.
- **Performance engineering:** GPU/accelerator profiling, Roofline and PMU analysis, CUDA Graphs, `torch.compile`, and memory-management optimization.
- **Kernel and hardware optimization:** CUDA, Triton, TileLang, fused attention/FFN kernels, low-precision GEMM, and cache-aware data layouts.
- **AI platform engineering:** Ray Data/Serve, KubeRay, Kubernetes scheduling, observability, and GPU-cluster resource optimization.

Selected Recognition

- IEEE AICAS 2026, Efficient VLM Inference and Optimization for AI Chips: **Rank 6 / 514**.
- vLLM-Kunlun open-source contributor: **Rank 6 / 39; 9 merged PRs** in framework, operator, and memory optimization.
- Shanghai Municipal and Aijian Special Foundation Scholarships; Outstanding Graduate and Youth League Member.